

MERGE

Master Product Requirements Document

Engineering Formation Platform for Nigerian CS Undergraduates

Version 1.0 · Semicolon Africa Capstone · Confidential

1. Product Overview

Merge is an Engineering Formation Platform for Nigerian CS undergraduates. It is not a bootcamp, an LMS, a coding platform, or a course marketplace. It transforms first-year university students into trusted software engineers through deliberate engineering formation over 3.5 years — running alongside the university degree, not replacing it.

Product Name	Merge
Type	Engineering Formation Platform
Primary User	David — 21-year-old Nigerian CS undergraduate, Lagos, 300 Level, mobile-first, 4 hours available on good days
Platform	React 18 PWA (frontend) + Java 21 Spring Boot (backend)
Architecture	Modular Monolith — scalable to microservices without rewrite
Database	Supabase PostgreSQL 15 + pgvector
AI	Google Gemini API (each student uses their own token)
Version	MVP — Scout + Cadet stages, 15 real students
Promise	"By the time you graduate, you're not entering the global tech market. You're already in it."

2. Problem Statement

REFINED PROBLEM STATEMENT

"How might we shift the mindsets of tech undergrads from optimizing for a degree to building genuine, industry-ready capability?"

METRIC	DATA	SOURCE
Graduate unemployability	60% of Nigerian graduates unemployable at point of hiring	Veriv Africa, 2024
Skills gap	Only 8.5% of CS graduates could explain SQL vs NoSQL in Lagos study	Ajuzieogu, 2025
Earnings gap	5-10x between local hire (N80-200k/month) and skilled remote dev (\$53,658/year)	Arc.dev, 2024
Brain drain	\$2.3B in human capital lost — Nigeria and Ghana — 2023	Dzreke et al., 2025
Skills-based hiring	81% of global employers now use skills-based hiring	TestGorilla, 2024
Funding probability	34/100 pre-MVP. Target 60/100 with working loop + 15 students	Internal

3. Stage Structure

Merge uses engineering-ranked stage names. David does not say "I am Level 2." He says "I am an Engineer on Merge." Five stages, each with a defined engineering question it answers.

STAGE	SEMESTER	ENGINEERING QUESTION	AI ACCESS	INTERNSHIP	XP THRES HOLD
SCOUT	Pre-Cadet	Who is David as a learner?	None	None	Auto on completion
CADET	1-2	Can David think before he codes?	None	None	1,500 XP + Build 600
ENGINEER	3-4	Can David build systems that survive reality?	Reviewer only	Local Nigerian	3,500 XP + Build 1,500
ARCHITECT	5-6	Can David make decisions under ambiguity?	Pair programmer	Global remote	7,000 XP + Build 3,000
PRINCIPAL	7	Can David be trusted with what matters?	Full accelerator	Full production	No ceiling

4. The Core Learning Loop

Every concept in every stage follows this exact sequence. No exceptions. The loop is the product.

1. Failure Scenario	David opens a concept. The first thing he sees is a real-world incident — named company, specific consequence, real financial or operational loss. Not hypothetical. He feels the consequence before touching the concept.
2. Merge Written Explanation	Merge's own engineering-grade written explanation of the concept. 500-800 words. Personalised by the AI based on David's learning approach from Scout. Not a YouTube link. Merge owns this content.
3. Resources Tab	Supplementary depth — specific book titles with exact chapters, YouTube timestamps, engineering papers. Always optional. Never required to pass.
4. Syntax Micro-Exercise	Cadet only. 5 minutes. Pattern recognition — fix broken code, fill keywords, translate pseudocode. Not assessed. Not XP. Builds automaticity so David's brain is free during the Drill.
5. Drill 1 — Guided	AI-generated PRD with architecture reasoning questions David must answer BEFORE submitting. Framed inside a MergeLabsCompany. Hints available in Cadet (tracked as weakness signals from Engineer onwards). TDD required.
6. Judge0 Execution	David's code runs against hidden test cases in an isolated sandbox. Network disabled. 2s CPU limit. 256MB memory. Pass or fail — no partial credit.

7. Comprehension Check	Triggers IMMEDIATELY on submission. Questions generated from David's specific code — his variable names, his function choices, his architectural decisions. 10-second timer enforced SERVER-SIDE. Cannot look anything up.
8. XP Award	Atomic database transaction. SELECT FOR UPDATE. Cap checked. Remainder awarded. Never from frontend. Always server-side.
9. Clean Code Feedback	Async. AI reviews against stage-appropriate rubric. Cadet: naming only. Engineer: +size+redundancy. Architect: full SOLID. Principal: senior human review.
10. Drill 2 — Independent	New scenario. No hints at any stage. Starts with mandatory code reading task. Same gate sequence. Same async jobs.
11. GitHub Auto-Commit	Async via BullMQ. Every approved submission commits to David's GitHub automatically. His contribution graph builds passively as a by-product of the grind.
12. Competency Evidence	Async. Every submission, comprehension check, and feedback event extracts competency signals tagged to one or more of the 8 competency dimensions.

5. The FACT Framework

Every concept in every stage uses FACT. Non-negotiable. What changes across stages is the depth and ambiguity — not whether it appears.

F — Failure First

What	Every concept is introduced through a real-world failure scenario. Real company. Real consequence. Real loss. Makes David feel the cost of ignorance before teaching the solution.
Escalation	Cadet: isolated, concrete — one thing broke. Engineer: cascading — one bad decision caused multiple failures. Architect: systemic — the entire design was wrong. Principal: preventable — a senior engineer reviewing the code would have caught it.

A — Architecture Reasoning

What	Every Drill PRD contains questions David must answer BEFORE submitting. He cannot submit code without justifying his design decisions. The comprehension check validates his answers.
Escalation	Cadet: "Why this approach? What if empty? What is the time complexity?" Engineer: trade-off comparison. Architect: defend against two alternatives. Principal: David generates his own architectural questions.

C — Code Reading

What	Every Drill 2 begins with a mandatory code reading task. David reads before he writes. This is the most neglected skill in Nigerian CS education and the most valued by employers.
Escalation	Cadet: 15-20 lines, spot one improvement. Engineer: 50-100 lines with real bug, find and fix. Architect: 200-500 lines of production module, critique architecture. Principal: real pull request, review as senior engineer.

T — TDD Always

What	Test-Driven Development is the default from Day 1 of Cadet to the last day of Principal. Red Green Refactor. Write a failing test first. No submission accepted without a test suite.
Escalation	Cadet: happy path + one edge case. Engineer: happy path + edge cases + error cases + boundary conditions. Architect: test suite reviewed as part of Build assessment. Principal: senior engineer reviews test suite in Merge Originals code review.

6. Engineering Competency Framework

Eight dimensions that measure engineering formation. Globally portable. Evidence-backed. SFIA-mapped for international legibility. Seeded once. Never changed. Evidence accumulates over 3.5 years.

#	COMPETENCY	WHAT IT MEASURES	PRIMARY EVIDENCE SOURCE	SFIA CATEGORY
1	Problem Solving	Breaking down and solving engineering problems	Architecture reasoning answers, Build decomposition	Systems Development
2	Software Design	Designing maintainable software and APIs	Build architecture documents, API design at Engineer+	Systems Design
3	Code Quality	Writing clean, readable, maintainable code	Clean Code feedback score trend across submissions	Programming / SWE
4	Testing & Debugging	Verifying behavior and diagnosing issues	TDD gate, test suite coverage, comprehension check	Testing
5	Systems Thinking	Understanding interactions between components	Engineer/Architect Builds, system design documents	Systems Architecture
6	Collaboration	Working effectively through reviews and communication	Peer review quality, Merge Labs contributions	Stakeholder Mgmt
7	Engineering Ownership	Taking responsibility for delivery and quality	Comprehension check scores, reflection records	Quality Management
8	Professional Growth	Reflection, adaptability, continuous improvement	Reflection outputs, scaffolding reduction over time	Learning & Development

7. XP System

XP is David's passport. Directional — every XP maps to a SFIA skill dimension. Two-gate promotion: minimum XP AND minimum Build Pass Score required simultaneously. XP capped per activity type per stage.

ACTIVITY	XP	CAP PER STAGE	NOTES
Learning resource completion	5 XP — once	50 XP	Curriculum resources only
Recap audio	0 XP — always	N/A	Memory anchor — never earns XP
Exhausted session audio	5 XP	—	Reinforcement or primer audio
Drill 1 — first attempt	25 XP	—	Decays on retry
Drill 2 — first attempt	50 XP	—	Decays on retry
Second attempt pass	75% of value	—	Encourages genuine preparation
Third attempt onwards	50% of value	—	Further decay
Weekly consistency 3+ sessions	+20 XP bonus	150 XP	Replaces daily streak
Weekly consistency 4/5 weekdays	+50 XP bonus	150 XP	—
Season top 25%	+100 XP	200 XP	Permanent badge awarded
Season top 10%	+200 XP	200 XP	Permanent gold badge
Build — test cases only	150 XP	150 XP	Minimum pass
Build — standard pass	300 XP	300 XP	Test cases + comprehension
Build — distinction (all 5 gates)	500 XP	500 XP	Maximum — all gates cleared
Peer review pass	30 XP	—	Architect stage only, spot-check verified
Merge Originals merged PR	200 XP	—	Principal stage only

8. Mood-Driven Session Model

The curriculum sequence is fixed and non-negotiable. What adapts is how the concept is delivered in any given session. David cannot commit to a consistent daily schedule — the platform accommodates his real life without compromising the curriculum.

MOOD	SESSION TYPE	CONTENT DELIVERED	XP
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FRESH	Full session	Failure scenario → Merge explanation → Resources → Syntax exercise (Cadet only) → Drill 1 → Drill 2	Full XP
OKAY	Standard session	Merge explanation → Drill 1 → Drill 2	Full XP
EXHAUSTED (mid-concept)	Reinforcement audio	AI generates 5-7 min audio on current concept from different angle. Seeds understanding. Drill still required next session.	5 XP
EXHAUSTED (concept complete)	Primer audio	AI generates 5-7 min audio priming the next concept. David arrives at next session already familiar with it.	5 XP

9. AI Architecture — Nine Specialist Prompt Roles

AI is not the product. AI is invisible infrastructure. The AI does not generate generic content — it generates David's content, calibrated to his 3.5 years of embedded history retrieved from pgvector before every Gemini call.

AI ROLE	TRIGGERED WHEN	CONTEXT FROM pgvector	OUTPUT
Curriculum Writer	Concept opened	Thinking style, learning approach, prior exposure, concept history	Personalised written explanation (500-800 words)
Drill Generator	Drill requested	Full Drill history, weakness signals, hint patterns, coding style	Unique PRD with architecture questions, MergeLabsCompany context
Comprehension Generator	Submission passes tests	David's specific code, previous check scores	Questions about his specific implementation, his variable names, his choices
Clean Code Reviewer	Comprehension passes	Previous feedback history, stage, current code	Specific named issues — escalating by stage. Naming → +size → +SOLID → human
Build PRD Generator	Build unlocked	Complete Drill performance history, weak/strong concepts, coding patterns	Personalised Build PRD targeting David's specific gaps
Reinforcement Audio	Exhausted + mid-concept	Current concept, struggle points from comprehension check	Script for 5-7 min audio covering concept from different angle
Primer Audio	Exhausted + concept complete	Next concept, David's learning approach	Script for 5-7 min audio seeding next concept
Reflection Assistant	Build or graduation complete	David's Build submission, stage journey, competency profile	Prompts that surface Engineering Identity evidence
Disengagement Coach	Momentum = BLOCKED/OFFLINE	Last activity, current concept, momentum history	Personalised reach-out message referencing David's actual journey

10. Security Requirements

RISK	PROTECTION	IMPLEMENTATION
AI cheating — comprehension bypass	Questions from David's specific code. 10s timer server-side. Different question each attempt.	ComprehensionCheck.serverDeadline enforced in backend. Timer rejection returns 400.
Token theft — Gemini token	Token never in frontend. Never in localStorage. Encrypted AES-256 in Supabase.	TokenEncryptionService. Backend makes all Gemini calls. Frontend never has token.
Code execution escape	Judge0 isolated Docker container. Network disabled. 2s CPU, 256MB memory.	ENABLE_NETWORK=false. cpu_time_limit=2. memory_limit=262144 (KB).
GitHub token abuse	Minimum scope: public repo write only. Encrypted at rest. Auto-revoke on suspicious activity.	OAuth scope: public_repo. Token encrypted same as Gemini token.
XP manipulation	XP never awarded from frontend. All server-side. SELECT FOR UPDATE atomic transactions.	XPService.awardXP() uses BEGIN TRANSACTION + optimistic locking.
Submission replay attack	Every submission has unique idempotency key. Replayed requests rejected 409.	submissions.idempotency_key UNIQUE constraint. Check before processing.
Peer review gaming	Senior engineer spot-checks. XP reversed on failure. Three failures suspend access.	PeerReviewService.spotCheck(). Async BullMQ job SPOT_CHECK_PEER_REVIEW.
Race conditions on XP	Optimistic locking. Database-level constraints. Atomic transaction.	SELECT total_xp FOR UPDATE. Check cap. Award or reject atomically. COMMIT.
JWT expiry mid-session	Silent refresh 5 minutes before expiry. Work saved to localStorage if refresh fails.	Frontend refresh interceptor. Restore unsaved work on next login.
Data privacy breach	Row-level security in Supabase. Students read only their own data. Companies see aggregated signals.	RLS policies on all student tables. auth.uid()::text = student_id::text

11. MVP Definition

The MVP has one job: prove the full loop closes with 15 real students. A student enters, the belief shifts, real capability builds, and an employer validates the output.

IN MVP

- ✓ Scout assessment — three layers
- ✓ Cadet stage — all 12 concepts, Drills, comprehension check, XP
- ✓ Cadet graduation assessment — Q&A; format
- ✓ Weekly Momentum Score — five states
- ✓ Cohort visibility — momentum states public
- ✓ GitHub auto-commit on every approved submission
- ✓ Basic student profile — stage, XP, momentum, competency status
- ✓ PWA configuration — offline passive content caching
- ✓ Merge written explanations — 12 Cadet concepts
- ✓ Drill generation via Gemini — David's own token

OUT OF MVP

- ✗ Engineer, Architect, Principal stages
- ✗ Internship platform
- ✗ Merge Originals and Gigs
- ✗ Company portal
- ✗ Peer review system
- ✗ Full AI personalisation engine — manual curation first
- ✗ Stripe / Pystack payments
- ✗ Algolia search
- ✗ Deepgram transcripts

SUCCESS METRICS

Metric 1	70% of students who complete Week 3 describe a belief shift when interviewed
Metric 2	80% of students who complete Cadet graduation assessment pass on first or second attempt
Metric 3	At least one company expresses genuine hiring interest after reviewing a student's Merge profile and GitHub